(Reaffirmed 0)

भारतीय मानक
Indian Standard

IS 4923 : 2017

खोखले इस्पात के खंड रचनात्मक प्रयोग के लिए — विशिष्टि

(तीसरा पुनरीक्षण)

Hollow Steel Sections for Structural Use — Specification

(Third Revision)

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omegaconsultant@gmail.com

ICS 77.140.75

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भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुरशाह ज़फर मार्ग, नई दिल्ली-110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI-110002
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FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Steel Tubes, Pipes and Fittings Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1968 and subsequently revised in 1985 and 1997. In view of growing demand of hollow steel sections for structural use, the committee decided to revise this standard incorporating the following main modifications:

a) Use of steel for welded tubes in line with IS 10748 has been incorporated

b) Table 1 has been modified to cover sizes of square hollow sections in line with International Standards

c) Table 2 has been modified to cover sizes of rectangular hollow sections in line with International Standards

d) Measurement of size and shape has been incorporated in line with International Standards.

e) A new YSt 355 Grade has been added in line with International Standards.

f) References have been updated

g) All 6 amendments have been incorporated

Assistance has been derived from the following:

i) ISO 10799-1 : 2011 Cold-formed welded structural hollow sections of non-alloy and fine grain steels — Part 1: Technical delivery conditions

ii) ISO 10799-2 : 2011 Cold formed welded structural hollow sections of non alloy and fine grain steels — Part 2 : Dimensions & sectional properties

iii) ISO 12633-1 : 2011 Hot-finished structural hollow sections of non-alloy and fine grain steels — Part 1: Technical delivery conditions

iv) BS EN 10219-1 : 2006 Cold formed welded structural hollow sections of non-alloy and fine grain steels. Technical delivery requirements

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be same as that of the specified value in this standard.

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Indian Standard

HOLLOW STEEL SECTIONS FOR STRUCTURAL USE — SPECIFICATION

(Third Revision)

1 SCOPE

This standard covers the requirements for hot and cold formed square and rectangular hollow steel sections for structural use

NOTE — Products manufactured to this specification may not be suitable for applications such as dynamically loaded elements in welded structures, where low-temperature notch toughness properties may be important.

2 REFERENCES

The following Indian Standards contain provisions, which through reference in this text, constitute provision of this standard. At the time of the publication, the editions indicated below were valid. All the standards are subject to revision, and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	*Title*
228	Methods of chemical analysis of pig iron, cast iron and plain carbon and low alloy steels
1387 : 1993	General requirements for the supply of metallurgical materials (*second revision*)
1608 : 2005	Mechanical testing of metals — Tensile testing (*second revision*)
10748 : 2004	Hot Rolled Steel strip for welded tubes and pipes (*second revision*)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Black Section

Section as manufactured, but without any subsequent surface treatment.

3.2 Exact Length

Specified length of a single section as mentioned by the purchaser.

3.3 Random Length

Normal manufacturing lengths which may vary over a range of several metres. Alternatively, a length range agreed to between the manufacturer and the purchaser.

3.4 Section

When used without qualification it refers to one length of square or rectangular hollow section covered by this specification.

4 SYMBOLS

For the purpose of this document, the symbols given below shall apply.

Symbol	Unit	Definition
A	Cm^2	Cross sectional area
B	mm	Nominal length of side of a square hollow section, Nominal length of shorter side of a rectangular hollow sectionor Width of section
$C1$, $C2$	mm	Length of external corner profile of a square or rectangular hollow section
e	mm	Deviation from straightness
L	mm	Length
H or D	mm	Nominal length of longer side of a rectangular hollow section or Depth of section
V	mm	Total twist
R	mm	External corner radius of square or rectangular hollow section
x_1	mm	Concavity of a side of square or rectangular hollow section
x_1	mm	Convexity of a side of square or rectangular hollow section
r	Degrees	Angle between adjacent sides of square or rectangular hollow section

SECTION 1 GENERAL REQUIREMENT

5 SUPPLY OF MATERIAL

5.1 General requirements relating to the supply of steel hollow sections shall conform to IS 1387.

6 DESIGNATION

6.1 A hollow section shall be designated by its outside

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dimensions and its thickness in millimeters and shall be further classified into CF or HF depending upon whether it is cold formed or hot formed.

Examples:

a) A hot formed square hollow section with outside dimensions of 50 mm square and 2.90 mm thickness is designated as 50 × 50 × 2.90 HF SHS.

b) A cold formed square hollow section with outside dimensions of 50 mm square and 2.90 mm thickness is designated as 50 × 50 × 2.90 CF SHS.

c) A hot formed rectangular hollow sectional with outside dimensions of 40 mm depth, 25 mm breadth and 2.65 mm thickness is designated as 40 × 25 × 2.65 HF RHS.

7 MANUFACTURE

7.1 The seamless hollow sections shall be manufactured from steel made by any approved process which shall show not more than 0.040 percent of Sulphur and 0.040 percent of Phosphorus. The welded hollow sections shall be manufactured from hot rolled steel strip conforming to IS 10748.

NOTE — If required the copper bearing steel may be used in manufacturing the hollow sections to achieve weather resistant properties subject to mutual agreement between the manufacturer and the purchaser. Copper content shall be between 0.20 to 0.35 percent.

7.1.1 The ladle analysis of the steel, shall be carried out by the method specified in the relevant parts of IS 228 or any other established instrumental/chemical method. In case of dispute the procedure given in IS 228 and its relevant parts shall be the referee method.

7.2 Tubes for forming into square and rectangular hollow sections shall be manufactured by one of the following processes:

a) Seamless,

b) Hot finished welded, or

c) Electric resistance or High Frequency Induction welded.

8. BASIS OF CALCULATION

8.1 For the purpose of the standard, the weights are calculated on the basis that steel weighs 0.785 kg/cm² per metre run. For calculating cross- sectional area and weight per unit length, corner radii have been taken into consideration (*see* Annex A).

9 DIMENSIONS AND WEIGHTS

9.1 The standard sizes and weights of hollow sections with other geometrical properties shall be as given in Tables 1 and 2. Sizes other than those given in Tables 1 and 2 may be supplied by mutual agreement between the supplier and the purchaser.

10. MESAUREMENT OF SIZE AND SHAPE

10.1 General

The tubes shall be supplied either in finish straightened or mill straightened condition as agreed to between the purchaser and the manufacturer for which maximum deviation from straightness shall be as per **14.1, 18.1.**

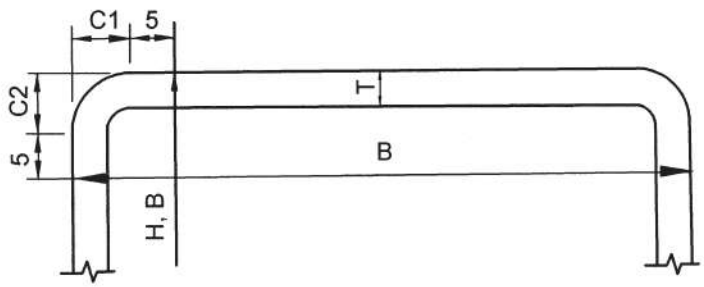
All external dimensions shall be measured at a distance from the end of the hollow section of not less than *B* for square sections or *H* for rectangular sections, with a minimum of 100 mm.

10.2 Outside Dimension

For square or rectangular hollow sections, outside dimensions, *B* & *H*, shall be measured. A vernier caliper gauge or other suitable device may be used at the discretion of the manufacturer.

Dimensions *B* and *H* shall be measured at a position within 5 mm from the start of the external corner profile as shown in Fig. 1. All Dimensions in millimeters.

This dimension is a maximum when measuring *B* or *H* and a minimum when measuring *T*.



All dimensions in millimetres.

FIG. 1 — LIMITING CROSS-SECTIONAL POSITIONS FOR MEASURING THE DIMENSIONS *B*, *H* AND *T* FOR SQUARE OR RECTANGULAR HOLLOW SECTIONS

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10.3 Thickness

The thickness, T, shall be measured at a position of not less than $2T$ from the weld. The limiting cross-sectional positions for measuring the thickness of square and rectangular hollow sections are shown in Figure 1.

NOTE — Thickness is normally measured within a distance of half the outside diameter or half the longer side length from the end of the section.

10.4 Concavity and Convexity

The concavity, x_1, or the convexity, x_2, of the sides of a square or rectangular hollow section shall be measured as shown in Fig. 2. The percentage concavity or convexity shall be calculated using the following:

$$\frac{x_1}{B}\times 100; \frac{x_2}{B}\times 100; \frac{x_1}{H}\times 100; \frac{x_2}{H}\times 100$$

*wh*ere B and H are the lengths of the sides containing the concavity, x_1, or the convexity, x_2.

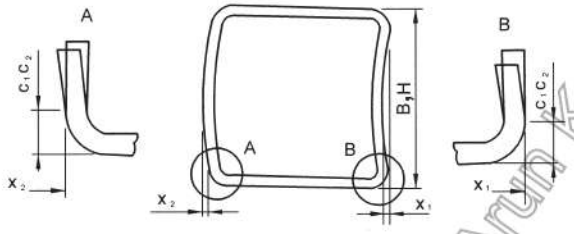
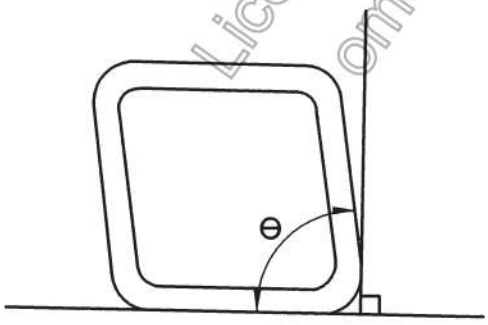


FIG. 2 MEASUREMENT OF CONCAVITY/CONVEXITY OF SQUARE OR RECTANGULAR HOLLOW SECTIONS

10.5 Squareness of Sides

The deviation from squareness of the sides of a square or rectangular hollow section is defined as the difference between 90° and θ as shown in Fig. 3.



Deviation from Squareness = 90 – θ

FIG. 3 SQUARENESS OF SIDES OF SQUARE OR RECTANGULAR HOLLOW SECTIONS

10.6 External Corner Radius

The external corner radius, R of a square or rectangular hollow section (Fig. 4) is measured by using radius gauge or other suitable device.

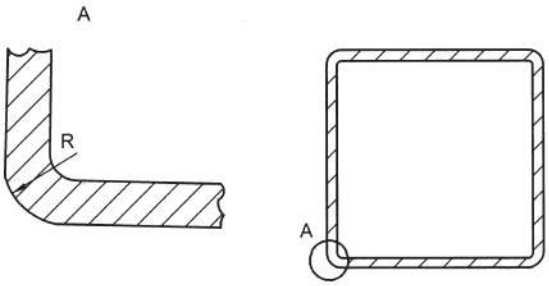


FIG. 4 EXTERNAL CORNER PROFILE OF SQUARE OR RECTANGULAR HOLLOW SECTIONS

10.7 Straightness and Twist

The total twist, V, in a square or rectangular hollow section shall be determined in accordance with any of the following methods, at the discretion of the manufacturer.

Place the hollow section on a horizontal surface with one side at one end pressed flat against the surface. At the opposite end of the hollow section, determine the difference, V_1, in the height of the two lower corners from a horizontal surface (*see* Fig. 5).

Measure V with a spirit level and micrometer (screw) gauge or other suitable device. The reference length of the spirit level shall be the distance between the intersection of the flat sides and the external corner profile (*see* Fig. 6). V is the difference between the values, V_1 (*see* Fig. 6), measured at each end of the section.

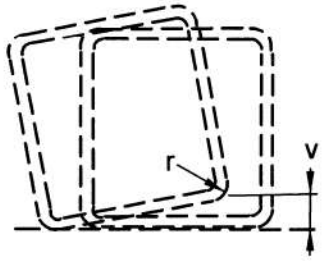
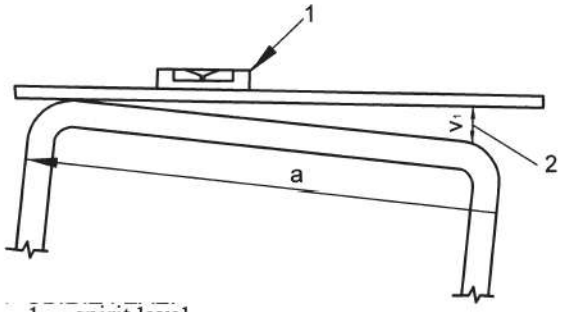


FIG. 5 TOTAL TWIST OF SQUARE OR RECTANGULAR HOLLOW SECTIONS KEY



1 spirit level
2 micrometer gauge
a H for rectangular sections, B for square sections.

FIG. 6 MEASUREMENT OF TWIST

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Twist shall be measured for square and rectangular sections as given in Fig. 1.

r_{max} = Maximum permitted outside comer radius in mm.

V = Total measured twist in millimetres.

11 OILING AND PAINTING

11.1 Hollow sections may be varnished, painted or oiled externally or may be galvanized as agreed between the purchaser and manufacturer. In case of galvanizing , the mass of zinc coating shall be as per the agreement between the manufacturer and purchaser. Pre galvanized sheet may be used for rolling hollow sections if so agreed between the purchaser and manufacturer.

12 PACKING

12.1 Where hollow sections are to be bundled for transport, these shall, unless otherwise specified, be secured together by rope, soft wire or straps. If other packing is required, it shall be agreed to between the purchaser and the manufacturer.

13 MARKING

13.1 Manufacturer name/Logo/Trade-mark shall be embossed on each hollow section. Each hollow section shall have size designation suitably marked on it. Alternatively, a label containing the particulars may be attached to a bundle of hollow sections.

13.1.1 Other particulars required by the purchaser may be suitably-marked as mutually agreed.

13.2 BIS Certification Marking

13.2.1 Hollow sections may also be marked with the Standard Mark.

13.2.2 The use of the Standard Mark is governed by the provisions of *Bureau of Indian Standard Act*,1986 and the rules and regulations made there under the details of conditions under which the licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

SECTION 2 SPECIFIC REQUIREMENTS FOR HOT FORMED SECTIONS

14 TOLERANCES

14.1 The following tolerances shall be permitted on hot formed hollow sections:

a)	Thickness for all sizes	
	1) Welded tubes	+ 7.5 Percent
	2) Seamless tubes	+ 12.5 Percent
b)	Outside dimensions of sides	+ 1 percent of length of the side to be measured with a minimum of + 0.5 mm
c)	Weight	+ 10 Percent
	1) On individual length	– 8 Percent
	2) On lots of 10 tonnes min.	+ 7 Percent
d)	Squareness of corner	$90^0 + 2^0$
(e)	Radii of corners— Outside	3t max where t is the thickness of section
f)	Length	
	1) Exact length	+ 6 mm
	2) Random length	This may be obtained by arrangement between purchaser and manufacturer
g)	Concavity/Convexity	1 percent of the size
h)	Twist	2 mm plus 0.5 mm/m measured in accordance with the figure
j)	Straightness	The tubes shall be supplied either in finish or mill straightened condition as agreed to between the purchaser and the manufacturer for which maximum deviation from straightness shall as under $1/600^{th}$ of length at the centre of length for finished straightened condition.

15 MECHANICAL TESTS

15.1 One tensile test shall be carried out for each batch of 500 hollow sections.

15.2 When tested in accordance with IS 1608, the tensile properties shall be as shown in Table 3.

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Elongation on a gauge length of $5.65\sqrt{A}$ (where, A is the cross-sectional area of test specimen) shall be as in Table 3.

Table 3 Tensile Properties of Hot Formed Sections
(*Clause* 15.2)

Grade	Yield Strength *Min*, MPa	Tensile Strength *Min*, MPa	Elongation *Min*, Percent
(1)	(2)	(3)	(4)
YSt 210	210	330	20
Yst 240	240	410	15
Yst 310	310	450	10
YSt 355	355	490	10

NOTE — For welded tubes, the strip tensile teat specimen shall not include the weld.

16 RETEST

16.1 Should any one of the test pieces selected fail to pass the test specified under 15, two further samples shall be selected for testing in respect of each failure. Should the test pieces from both these additional samples pass, the material shall be deemed to comply with the requirements of the test. Should the test pieces from either of these additional samples fail, the material represented by the test samples shall be deemed as not complying with this standard.

SECTION 3 SPECIFIC REQUIREMENTS FOR COLD FORMED SECTIONS

17 CONDITION OF SUPPLY

17.1 The section shall be supplied in as welded condition. The section may also be stress relieved, annealed or normalized as agreed between manufacturer and purchaser.

18 TOLERANCES

18.1 The following tolerances shall be permitted on cold formed hollow sections:

a)	Thickness for all sizes	
	1) Welded tubes	+ 7.5 Percent
	2) Seamless tubes	+ 12.5 Percent
b)	Outside dimensions	+ 1 percent of length of the side to be measured with a minimum tolerance of + 0.5 mm
c)	Weight	+ 10 Percent
	1) On individual length	– 8 Percent
	2) On lots of 10 tonnes, Min	+ 7 Percent
d)	Squareness of corner	90°+ 2°
e)	Radii of corners— Outside	3t max where t is the thickness of section
f)	Length	
	1) Exact length	+ 6 mm
	2) Random length	This may be obtained by arrangement between purchaser and manufacturer
g)	Concavity/Convexity	1 percent of the size
h)	Twist	2 mm plus 0.5 mm/m measured in accordance with the figure
j)	Straightness	The tubes shall be supplied either in finish or mill straightened condition as agreed to between the purchaser and the manufacturer for which maximum deviation from straightness shall as under 1/600th of length at the centre of length for finished straightened condition. For tubes in mill straightened condition 1/200th of any length at the centre of the length.

NOTE

i) The measurement of thickness should exclude the weld zone

ii) The height of the internal weld fin shall not exceed 60% of the wall thickness

19 MECHANICAL TESTS

19.1 One tensile test shall be carried out for each of 500 hollow sections.

19.2 When tested in accordance with IS 1608, the tensile properties of cold formed sections and elongation percentage on a gauge length of $5.65\sqrt{A}$ (where, A is the cross sectional area of the section) shall be as in Table 4.

NOTE — The tensile test piece shall be taken excluding the side of the section on which the weld seam exists.

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Table 4 Tensile Properties of Cold Formed Sections

(*Clause* 19.2)

Grade	Yield Strength *Min*, MPa	Tensile Strength *Min*, MPa	Elongation Percent *Min*, for Sizes	
			25.4 and under *	Over 25.4 *
(1)	(2)	(3)	(4)	(5)
YSt 210	210	330	12	20
Yst 240	240	410	10	15
Yst 310	310	450	8	10
YSt 355	355	490	8	10

NOTE — i) *The value shall be applicable for the smaller side of the rectangular section.

ii) Stress relief annealing at more than 580°C or for over one hour may lead to deterioration of the mechanical properties.

20 RETEST

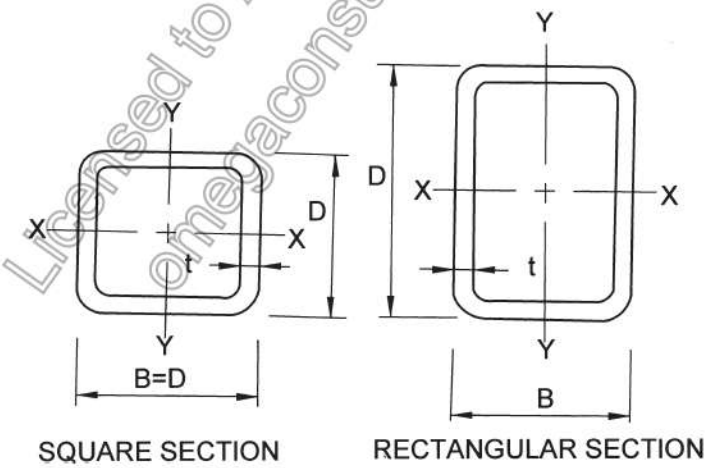
20.1 Should any of the pieces selected fail to pass the test specified under 19, two further samples shall be selected for testing in respect of each failure. Should the test pieces from both of these additional samples pass, the material shall be deemed to comply with the requirements of the test. Should the test pieces from either of these additional samples fail, the material represented by the test samples shall be deemed as not complying with this standard.

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ANNEX A

(*Clause* 8.1)

GEOMETRICAL DATA OF HOLLOW SECTION



Area of cross-section = A = $2t\left[(B-4t)+(D-4t)+\frac{3}{2}\pi t\right]$ in cm^2

Weight (kg/m) = W = 0.785 A in cm^2

For X–X axis:

Moment of incrtia = Ix = $t\frac{(D-4t)^3}{\sigma}+\frac{1}{2}\left[\frac{(B-4t)t^3}{3}+(B-4t)(D-r)^2{}_t\right]+$

$$\frac{\pi t^4}{108}\left[405-\frac{3136}{\pi^2}\right]+3\pi t^2\left[\frac{9\pi(D-4t)+56t}{18\pi}\right] \text{ in cm}^4$$

Elastic modulus $= Zx = \frac{2I_x}{D}$ in cm^3

Plastic modulus $= Sx = t/2\ (D - 4t)2 + t(B-4t)(D-t) + \frac{t^2}{\sigma}\left[9\pi\left(D-4t+56t\right)\right]$ in cm^3

Radius of gyration $= Rx = \sqrt{\frac{I_x}{A}}$ in cm

For Y-Y axis:

Moment of inertia $= Ix = + \frac{\pi t^4}{108}\left[405-\frac{3136}{\pi^2}\right]+3\pi t^2\left[\frac{9\pi\left(D-4t\right)+56t}{18\pi}\right]$ in cm^4

Elastic modulus $= Zx = \frac{2I_x}{D}$ in cm^3

Radius of gyration $= Ry = \sqrt{\frac{I_x}{A}}$ in cm

For Y –Y axis:

Moment of inertia $= Iy = t\frac{\left(B-4t\right)^3}{6}+\frac{1}{2}\left[\frac{\left(D-4t\right)t^3}{3}+\left(D-4t\right)\left(B-r\right)^2 t\right]+$

$\frac{\pi t^4}{108}\left[405-\frac{3136}{\pi^2}\right]+3\pi t^2\left[\frac{9\pi\left(B-4t\right)+56t}{18\pi}\right]$ in cm^4

Elastic modulus $= Zy = \frac{2I_x}{B}$ in cm^3

Plastic modulus $= Sy = t/2\ (B - 4t)^2 + t(D - 4t)\ (B - t) + \frac{t^2}{\sigma}\left[9\pi\left(B-4t+56t\right)\right]$ in cm^3

Radius of gyration $= Ry = \sqrt{\frac{I_x}{A}}$ in cm

Note — Letter symbols denoting various dimensions are the same as those used in Tables 1 and 2.

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Table 1 Dimensions and Properties of Square Hollow Sections

(*Clause* 9.1 *and Note in Annex A*)

Designation/ Size mm (1)	Depth *D* mm (2)	Width *B* mm (3)	Thickness *t* mm (4)	Weigth kg/m (5)	Area of Section cm^2 (6)	Moment of Inertia cm^4 (7)	Radius of Gyration cm (8)	Elastic Modulus cm^3 (9)	Plastic Modulus cm^3 (10)
25 × 25 × 2	25	25	2.00	1.36	1.74	1.48	0.92	1.19	1.47
25 × 25 × 2.6	25	25	2.60	1.69	2.16	1.72	0.89	1.38	1.76
25 × 25 × 3.2	25	25	3.20	1.98	2.53	1.89	0.86	1.51	1.98
30 × 30 × 2	30	30	2.00	1.68	2.14	2.72	1.13	1.81	2.21
30 × 30 × 2.6	30	30	2.60	2.10	2.68	3.23	1.10	2.15	2.68
30 × 30 × 3.2	30	30	3.20	2.49	3.17	3.62	1.07	2.41	3.08
32 × 32 × 2.6	32	32	2.60	2.26	2.88	4.02	1.18	2.51	3.11
32 × 32 × 3.2	32	32	3.20	2.69	3.42	4.54	1.15	2.84	3.59
32 × 32 × 4	32	32	4.00	3.19	4.07	5.02	1.11	3.14	4.11
35 × 35 × 2.6	35	35	2.60	2.51	3.20	5.43	1.30	3.10	3.81
35 × 35 × 3.2	35	35	3.20	2.99	3.81	6.18	1.27	3.53	4.42
35 × 35 × 4	35	35	4.00	3.57	4.55	6.93	1.23	3.96	5.11
38 × 38 × 2	38	38	2.00	2.18	2.78	5.88	1.46	3.10	3.70
38 × 38 × 2.6	38	38	2.60	2.75	3.51	7.14	1.43	3.76	4.57
38 × 38 × 3.2	38	38	3.20	3.29	4.19	8.18	1.40	4.30	5.34
38 × 38 × 4	38	38	4.00	3.95	5.03	9.26	1.36	4.87	6.22
40 × 40 × 2	40	40	2.00	2.31	2.94	6.94	1.54	3.47	4.13
40 × 40 × 2.6	40	40	2.60	2.92	3.72	8.45	1.51	4.22	5.12
40 × 40 × 3.2	40	40	3.20	3.49	4.45	9.72	1.48	4.86	6.00
40 × 40 × 4	40	40	4.00	4.20	5.35	11.07	1.44	5.54	7.01
45 × 45 × 2.6	45	45	2.60	3.33	4.24	12.42	1.71	5.52	6.64
45 × 45 × 2.9	45	45	2.90	3.66	4.67	13.45	1.70	5.98	7.25
45 × 45 × 3.2	45	45	3.20	3.99	5.09	14.41	1.68	6.40	7.82
45 × 45 × 4	45	45	4.00	4.83	6.15	16.61	1.64	7.38	9.22
45 × 45 × 4.5	45	45	4.50	5.31	6.77	17.74	1.62	7.88	9.99
49.5 × 49.5 × 2.6	49.5	49.5	2.60	3.69	4.70	16.91	1.90	6.83	8.16
49.5 × 49.5 × 3.2	49.5	49.5	3.20	4.45	5.66	19.74	1.87	7.98	9.67
49.5 × 49.5 × 4	49.5	49.5	4.00	5.39	6.87	22.95	1.83	9.27	11.46
63.5 × 63.5 × 3.2	63.5	63.5	3.20	5.85	7.46	44.35	2.44	13.97	16.65
63.5 × 63.5 × 3.6	63.5	63.5	3.60	6.51	8.29	48.55	2.42	15.29	18.36
63.5 × 63.5 × 4.5	63.5	63.5	4.50	7.93	10.10	57.00	2.38	17.95	21.93
72 × 72 × 3.2	72	72	3.20	6.71	8.54	66.32	2.79	18.42	21.80
72 × 72 × 4	72	72	4.00	8.22	10.47	79.03	2.75	21.95	26.32
72 × 72 × 4.8	72	72	4.80	9.66	12.31	90.31	2.71	25.09	30.48
75 × 75 × 3.2	75	75	3.20	7.01	8.93	75.53	2.91	20.14	23.79
75 × 75 × 3.6	75	75	3.60	7.81	9.95	83.06	2.89	22.15	26.32
75 × 75 × 4.5	75	75	4.50	9.55	12.17	98.55	2.85	26.28	31.68
88.9 × 88.9 × 3.2	88.9	88.9	3.20	8.40	10.71	129.40	3.48	29.11	34.11
88.9 × 88.9 × 3.6	88.9	88.9	3.60	9.38	11.95	142.83	3.46	32.13	37.85
88.9 × 88.9 × 4.5	88.9	88.9	4.50	11.52	14.67	170.97	3.41	38.46	45.85
91.5 × 91.5 × 3.6	91.5	91.5	3.60	9.67	12.32	156.49	3.56	34.21	40.24
91.5 × 91.5 × 4.5	91.5	91.5	4.50	11.88	15.14	187.57	3.52	41.00	48.79
91.5 × 91.5 × 4.8	91.5	91.5	4.80	12.60	16.05	197.27	3.51	43.12	51.51
100 × 100 × 4	100	100	4.00	11.73	14.95	226.35	3.89	45.27	53.30
100 × 100 × 5	100	100	5.00	14.41	18.36	271.10	3.84	54.22	64.59
100 × 100 × 6	100	100	6.00	16.98	21.63	311.47	3.79	62.29	75.09
113.5 × 113.5 × 4.5	113.5	113.5	4.50	14.99	19.10	372.88	4.42	65.71	77.32
113.5 × 113.5 × 4.8	113.5	113.5	4.80	15.92	20.28	393.30	4.40	69.30	81.81
113.5 × 113.5 × 5.4	113.5	113.5	5.40	17.74	22.60	432.58	4.38	76.23	90.54
113.5 × 113.5 × 6	113.5	113.5	6.00	19.53	24.87	469.81	4.35	82.79	98.95
125 × 125 × 4.5	125	125	4.50	16.62	21.17	505.83	4.89	80.93	94.84
125 × 125 × 5	125	125	5.00	18.34	23.36	552.62	4.86	88.42	104.09
125 × 125 × 6	125	125	6.00	21.69	27.63	640.89	4.82	102.54	121.86
132 × 132 × 4.5	132	132	4.50	17.61	22.43	600.45	5.17	90.98	106.37
132 × 132 × 4.8	132	132	4.80	18.71	23.83	634.39	5.16	96.12	112.68
132 × 132 × 5.4	132	132	5.40	20.88	26.60	700.11	5.13	106.08	125.01
132 × 132 × 6	132	132	6.00	23.01	29.31	762.98	5.10	115.60	136.97

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Table 1 — (*Concluded*)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
150 × 150 × 5	150	150	5.00	22.26	28.36	982.12	5.89	130.95	152.97
150 × 150 × 6	150	150	6.00	26.40	33.63	1145.91	5.84	152.79	179.87
150 × 150 × 7	150	150	7.00	30.44	38.78	1299.44	5.79	173.26	205.57
150 × 150 × 8	150	150	8.00	34.38	43.79	1443.00	5.74	192.40	230.09
180 × 180 × 4	180	180	4.00	21.90	27.90	1434.00	7.17	159.00	184.00
180 × 180 × 5	180	180	5.00	27.20	34.60	1755.00	7.12	195.00	226.00
180 × 180 × 6	180	180	6.00	32.05	40.83	2036.00	7.06	226.00	280.00
180 × 180 × 8	180	180	8.00	42.50	54.10	2633.00	6.98	293.00	346.00
220 × 220 × 6	220	220	6.00	39.59	50.43	3813.36	8.70	346.67	402.16
220 × 220 × 8	220	220	8.00	51.96	66.19	4894.99	8.60	445.00	521.67
220 × 220 × 10	220	220	10.00	63.92	81.43	5887.19	8.50	535.20	634.11
220 × 220 × 12	220	220	12.00	75.47	96.14	6793.08	8.41	617.55	739.60
250 × 250 × 6	250	250	6.00	45.24	57.63	5672.00	9.92	453.76	524.43
250 × 250 × 8	250	250	8.00	59.50	75.79	7315.65	9.82	585.25	682.64
250 × 250 × 10	250	250	10.00	73.34	93.43	8841.86	9.73	707.35	832.74
250 × 250 × 12	250	250	12.00	86.77	110.54	10254.21	9.63	820.34	974.87
300 × 300 × 6	300	300	6.00	54.66	69.63	9963.67	11.96	664.24	764.21
300 × 300 × 8	300	300	8.00	72.06	91.79	12925.07	11.87	861.67	998.91
300 × 300 × 10	300	300	10.00	89.04	113.43	15713.90	11.77	1047.59	1223.79
300 × 300 × 12	300	300	12.00	105.61	134.54	18334.49	11.67	1222.30	1438.98
350 × 350 × 8	350	350	8.00	84.62	107.79	20849.89	13.91	1191.42	1375.18
350 × 350 × 10	350	350	10.00	104.74	133.43	25453.75	13.81	1454.50	1689.84
350 × 350 × 12	350	350	12.00	124.45	158.54	29824.41	13.72	1704.25	1993.09
400 × 400 × 8	400	400	8.00	97.18	123.79	31490.11	15.95	1574.51	1811.45
400 × 400 × 10	400	400	10.00	120.44	153.43	38561.41	15.85	1928.07	2230.89
400 × 400 × 12	400	400	12.00	143.29	182.54	45323.97	15.76	2266.20	2637.20

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Table 2 Dimensions and Properties of Rectangular Hollow Section

(*Clause* 9.1 *and* Note in Annex A)

Designation/ Size	Depth	Width	Thickness	Weight	Area of Section	Moment of Inertia about		Radius of Gyration about		Elastic Modulus about		Plastic Modulus about	
	D	*B*	*t*			X-X	Y-Y	X-X	Y-Y	X-X	Y-Y	X-X	Y-Y
mm	mm	mm	mm	cm^2	cm^3	cm^4	cm	cm	cm	cm^3	cm^3	cm^3	cm^3
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
50 × 25 × 2.9	50	25	2.90	2.98	3.80	10.93	3.60	1.70	0.97	4.37	2.88	5.72	3.48
50 × 25 × 3.2	50	25	3.20	3.24	4.13	11.63	3.80	1.68	0.96	4.65	3.04	6.14	3.73
60 × 40 × 2.9	60	40	2.90	4.12	5.25	24.74	13.11	2.17	1.58	8.25	6.56	10.24	7.73
66 × 33 × 2.6	66	33	2.60	3.69	4.70	25.15	8.43	2.31	1.34	7.62	5.11	9.68	5.94
66 × 33 × 2.9	66	33	2.90	4.07	5.19	27.33	9.12	2.29	1.33	8.28	5.53	10.59	6.49
66 × 33 × 3.6	66	33	3.60	4.93	6.28	31.87	10.52	2.25	1.29	9.66	6.37	12.56	7.65
66 × 33 × 4.5	66	33	4.50	5.95	7.58	36.64	11.93	2.20	1.25	11.10	7.23	14.76	8.94
70 × 30 × 2.9	70	30	2.90	4.12	5.25	29.82	7.72	2.38	1.21	8.52	5.14	11.07	6.04
70 × 30 × 3.2	70	30	3.20	4.50	5.73	32.04	8.24	2.37	1.20	9.15	5.49	11.98	6.51
70 × 30 × 4	70	30	4.00	5.45	6.95	37.23	9.42	2.31	1.16	10.64	6.28	14.19	7.66
80 × 40 × 2.9	80	40	2.90	5.03	6.41	50.87	17.11	2.82	1.63	12.72	8.56	16.07	9.88
80 × 40 × 3.2	80	40	3.20	5.50	7.01	54.94	18.41	2.80	1.62	13.74	9.21	17.46	10.71
80 × 40 × 4	80	40	4.00	6.71	8.55	64.79	21.49	2.75	1.59	16.20	10.74	20.91	12.77
96 × 48 × 3.2	96	48	3.20	6.71	8.54	98.61	33.28	3.40	1.97	20.54	13.87	25.85	15.91
96 × 48 × 4	96	48	4.00	8.22	10.47	117.54	39.32	3.35	1.94	24.49	16.38	31.20	19.14
96 × 48 × 4.8	96	48	4.80	9.66	12.31	134.35	44.55	3.30	1.90	27.99	18.56	36.13	22.07
100 × 50 × 3.2	100	50	3.20	7.01	8.93	112.29	37.95	3.55	2.06	22.46	15.18	28.20	17.37
100 × 50 × 4	100	50	4.00	8.59	10.95	134.14	44.95	3.50	2.03	26.83	17.98	34.10	20.93
122 × 6.1 × 3.6	122	61	3.60	9.67	12.32	232.61	78.83	4.34	2.53	38.13	25.84	47.71	29.42
122 × 6.1 × 4.5	122	61	4.50	11.88	15.14	278.94	93.78	4.29	2.49	45.73	30.75	57.84	35.55
122 × 6.1 × 5.4	122	61	5.40	14.01	17.85	320.83	107.03	4.24	2.45	52.60	35.09	67.28	41.22
127 × 50 × 3.6	127	50	3.60	9.34	11.89	227.08	52.05	4.37	2.09	35.76	20.82	45.95	23.70
127 × 50 × 4.6	127	50	4.60	11.69	14.89	276.33	62.46	4.31	2.05	43.52	24.98	56.65	29.03
145 × 82 × 4.8	145	82	4.80	15.92	20.28	555.16	228.50	5.23	3.36	76.57	55.73	94.93	63.93
145 × 82 × 5.4	145	82	5.40	17.74	22.60	610.85	250.59	5.20	3.33	84.26	61.12	105.06	70.66
172 × 92 × 4.8	172	92	4.80	18.71	23.83	917.13	346.91	6.20	3.82	106.64	75.41	132.07	85.61
172 × 92 × 5.4	172	92	5.40	20.88	26.60	1 012.47	381.74	6.17	3.79	117.73	82.99	146.54	94.85
200 × 100 × 4	200	100	4.00	18.01	22.95	1 199.71	410.78	7.23	4.23	119.97	82.16	148.03	91.70
200 × 100 × 5	200	100	5.00	22.26	28.36	1 459.25	496.94	7.17	4.19	145.93	99.39	181.36	112.09
200 × 100 × 6	200	100	6.00	26.40	33.63	1 703.31	576.91	7.12	4.14	170.33	115.38	213.25	131.49
200 × 100 × 8	200	100	8.00	34.38	43.79	2 146.21	719.19	7.00	4.05	214.62	143.84	272.76	167.42
220 × 140 × 4	220	140	4.00	21.78	27.75	1 892.55	947.64	8.26	5.84	172.05	135.38	206.34	151.67

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Table 2 — (*Concluded*)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
220 × 140 × 5	220	140	5.00	26.97	34.36	2 313.36	1 155.23	8.21	5.80	210.31	165.03	253.72	186.30
220 × 140 × 6	220	140	6.00	32.05	40.83	2 713.97	1 351.66	8.15	5.75	246.72	193.09	299.44	219.64
220 × 140 × 8	220	140	8.00	41.91	53.39	3 456.10	1 712.17	8.05	5.66	314.19	244.60	385.99	282.52
240 × 120 × 4	240	120	4.00	21.78	27.75	2 110.72	725.35	8.72	5.11	175.89	120.89	216.01	134.00
240 × 120 × 5	240	120	5.00	26.97	34.36	2 579.67	882.47	8.67	5.07	214.97	147.08	265.57	164.44
240 × 120 × 6	240	120	6.00	32.05	40.83	3 025.91	1 030.45	8.61	5.02	252.16	171.74	313.39	193.69
240 × 120 × 8	240	120	8.00	41.91	53.39	3 851.84	1 299.95	8.49	4.93	320.99	216.66	403.86	248.65
260 × 180 × 6	260	180	6.00	39.59	50.43	4 855.87	2 763.43	9.81	7.40	373.53	307.05	446.86	347.86
260 × 180 × 8	260	180	8.00	51.96	66.19	6 238.69	3 538.10	9.71	7.31	479.90	393.12	579.81	450.74
260 × 180 × 10	260	180	10.00	63.92	81.43	7 509.51	4 244.26	9.60	7.22	577.65	471.58	704.95	547.27
260 × 180 × 12	260	180	12.00	75.47	96.14	8 671.80	4 884.69	9.50	7.13	667.06	542.74	822.41	637.59
300 × 150 × 6	300	150	6.00	40.53	51.63	6 073.51	2 079.57	10.85	6.35	404.90	277.28	499.61	309.47
300 × 150 × 8	300	150	8.00	53.22	67.79	7 807.95	2 654.12	10.73	6.26	520.53	353.88	648.51	400.49
300 × 150 × 10	300	150	10.00	65.49	83.43	9 403.90	3 173.71	10.62	6.17	626.93	423.16	788.79	485.64
300 × 150 × 12	300	150	12.00	77.35	98.54	10 865.21	3 640.89	10.50	6.08	724.35	485.45	920.58	565.04
300 × 200 × 6	300	200	6.00	45.24	57.63	7 370.23	3 962.19	11.31	8.29	491.35	396.22	587.81	446.05
300 × 200 × 8	300	200	8.00	59.50	75.79	9 513.66	5 097.04	11.20	8.20	634.24	509.70	765.31	579.96
300 × 200 × 10	300	200	10.00	73.34	93.43	11 507.24	6 144.30	11.10	8.11	767.15	614.43	933.79	706.69
300 × 200 × 12	300	200	12.00	86.77	110.54	13 354.97	7 107.11	10.99	8.02	890.33	710.71	1 093.38	826.36
350 × 250 × 6	350	250	6.00	54.66	69.63	12 457.31	7 458.44	13.38	10.35	711.85	596.68	842.58	670.83
350 × 250 × 8	350	250	8.00	72.06	91.79	16 170.48	9 659.06	13.27	10.26	924.03	772.73	1 101.58	876.24
350 × 250 × 10	350	250	10.00	89.04	113.43	19 672.08	11 723.53	13.17	10.17	1 124.12	937.88	1 349.84	1 072.74
350 × 250 × 12	350	250	12.00	105.61	134.54	22 966.89	13 655.73	13.07	10.07	1 312.39	1 092.46	1 587.49	1 260.47
400 × 200 × 8	400	200	8.00	72.06	91.79	19 195.28	6 572.45	14.46	8.46	959.76	657.25	1 184.25	733.56
400 × 200 × 10	400	200	10.00	89.04	113.43	23 348.08	7 950.97	14.35	8.37	1 167.40	795.10	1 450.89	896.69
400 × 200 × 12	400	200	12.00	105.61	134.54	27 252.93	9 230.63	14.23	8.28	1 362.65	923.06	1 706.00	1 051.96
400 × 300 × 8	400	300	8.00	84.62	107.79	25 342.70	16 336.49	15.33	12.31	1 267.13	1 089.10	1 497.85	1 232.51
400 × 300 × 10	400	300	10.00	104.74	133.43	30 954.74	19 920.57	15.23	12.22	1 547.74	1 328.04	1 840.89	1 513.79
400 × 300 × 12	400	300	12.00	124.45	158.54	36 288.45	23 314.01	15.13	12.13	1 814.42	1 554.27	2 171.60	1 784.58
500 × 300 × 8	500	300	8.00	97.18	123.79	43 149.92	19 747.90	18.67	12.63	1 726.00	1 316.53	2 076.80	1 466.11
500 × 300 × 10	500	300	10.00	120.44	153.43	52 866.82	24 127.24	18.56	12.54	2 114.67	1 608.48	2 557.99	1 803.79
500 × 300 × 12	500	300	12.00	143.29	182.54	62 169.00	28 293.53	18.45	12.45	2 486.76	1 886.24	3 024.23	2 130.18

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